



Jeffrin Sam

✉ jeffrin.s.a@gmail.com ☎ +7 915 046 7795 📍 Moscow, Russia

🌐 LinkedIn 🔄 GitHub 🇮🇳 Indian 🔗 @JeffrinSam

🤖 huggingface.co/JeffrinSam 📄 scholar.google.com/citations

Profile

Robotics AI researcher at Skoltech specializing in embodied AI, Simulation, and control systems. Deep expertise in Vision-Language-Action (VLA) architectures, World Model, and video generation models—capable of designing, implementing, and training models from the ground up. Hands-on experience deploying VLA models and VLMs on Unitree H1/G1 humanoids and UAV systems in real-world environments, with active research in video generation (Cosmos 2.5, Wan 2.2) for robotics. Currently developing a unified framework combining task planning, VLA execution, and video-guided learning for humanoid control—when the robot fails, it learns from generated demonstration videos. Proficient in Python, C++, ROS2, MATLAB/Simulink, and NVIDIA Isaac Sim/Lab. Seeking Research Engineer or PhD opportunities in Robotics and AI to advance general-purpose cognitive robots through foundation world models and large-scale learning.

Professional Experience

10.2024 – Present

Graduate Student Researcher (Physical AI)

Skoltech, ISR Lab – Moscow 

Conduct research in Physical AI, Agentic AI, Vision-Language-Action (VLA) systems, and Foundation World Models for humanoid robots and autonomous systems. Develop large-scale synthetic data generation pipelines in **NVIDIA Isaac Sim/Lab for Unitree G1/H1**  humanoid robots, supporting imitation learning, reinforcement learning, and VLA training. Train, fine-tune, and evaluate robotics foundation models, including NVIDIA GRooT, **Cosmos** , and Wan 2.2, on multi-GPU cluster environments for robot learning, video generation, future-state prediction, and embodied intelligence applications. Design agentic architectures utilizing multi-agent systems and autonomous planning frameworks to improve robotic reasoning, task execution, and decision-making. Build simulation environments in Isaac Sim/Lab, MuJoCo, and Unity for humanoid locomotion, manipulation, sim-to-real transfer, and **digital twin development** 

09.2025 – 01.2026

Guest Instructor – Robotics Simulation

Yerevan State University 

Designed and delivered a comprehensive, **3-month advanced workshop** on robotics simulation. The curriculum provided hands-on experience with **NVIDIA Isaac Sim/Lab** and **Unity**, covering **ROS2 deployment**, **synthetic data generation**, and the implementation of **Reinforcement & Imitation Learning** methods for training robust robotic policies

05.2025 – 12.2025

Robotic Simulation Researcher

MTC Group 

Develop and validate Vision-Language-Action (VLA) models for bimanual robotic manipulation. My work involves creating simulations in NVIDIA Isaac Sim, collecting real-world data and simulation data, and fine-tuning models like **NVIDIA Isaac GRooT N1.5**  to automate data center maintenance tasks.

06.2023 – 02.2025

STEM Tutor & Project Advisor (Engineering & Analytics)

Al Manal Training L.L.C- Abu Dhabi, UAE 

I provide one-on-one and group tutoring sessions in Engineering Chemistry, Physics, Mathematics, Circuit Analysis, Electronic Devices and Circuits, Robotics and Computer Vision, Linear Algebra, Partial Differential Equations, Finite Calculus, Statistics, Business Analytics, Engineering Mathematics, Dynamics, and Kinematics. I communicate complex engineering concepts easily, enabling students to comprehend and retain information effectively. Additionally, I mentor students on final-year **projects and Work on innovative ideas**, and apply theoretical knowledge to practical applications

07.2023 – 08.2024

Robotic Instructor

Time Master-Abu Dhabi, UAE 

Developed and taught a dynamic curriculum for teenagers using the RoboRobo module, focusing on hands-on robot assembly, electronics, and programming in Scratch and C++. Designed multi-level courses covering 6V/12V systems, Bluetooth technology, and logic programming, integrating electrical principles. Fostered students' creativity, logic, and problem-solving skills, preparing them for university success.

06.2021 – 01.2023

Production Engineer

ARASAN SOAP WORKS PLC :IN-629193

In this role, I oversaw production processes and managed machinery maintenance while analyzing power systems using MATLAB/Simulink to enhance operational efficiency. I also led the "**Decarbonization of Industry**" project, which focused on integrating sustainable practices and renewable energy systems into our operations.

Projects

05.2025 – 12.2025

NVIDIA Isaac Lab for VLA and Imitation Learning with Unitree Robots

Using NVIDIA Isaac Lab, I generate synthetic data for imitation learning by teleoperating Unitree G1 and H1 robots in simulated environments, capturing diverse demonstrations for tasks like pick-and-place. These demonstrations are annotated and augmented with Isaac Lab Mimic to create large-scale datasets, which are used to train Vision-Language-Action (VLA) models. The trained policies are deployed in Isaac Sim, with visuals enhanced via Cosmos for realism, enabling robust policy training for G1 and H1, including potential fine-tuning for models like GRooT.

10.2024 – 12.2024

Concept of Prototype-LIDAR sensing for CubeSat

This project explores the concept of a prototype LIDAR sensing system for a CubeSat, utilizing a 2D LIDAR sensor to estimate ice thickness and map icebergs. The concept involves analyzing LIDAR data and comparing it with other sources, including ICESat-2 satellite data, to assess feasibility and potential accuracy. The study focuses on conceptual design, data analysis methodologies, and possible implementation strategies for future CubeSat missions

2022 – 2023

Decarbonization of Industry- 30KW

Implemented an eco-friendly solution by substituting fuel wood with a thermal electric boiler powered by solar energy for boiling and incorporating a solar-powered motor for mixers, aiming to significantly reduce the carbon footprint in the detergent industry and promote the use of renewable energy.



Publications

Action Agent: Agentic Video Generation Meets Flow-Constrained Diffusion

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)

Accepted will be published in Pittsburgh, USA

SafeHumanoid: VLM-RAG-driven Control of Upper Body Impedance for Humanoid Robot

ACM/IEEE International Conference on Human-Robot Interaction (HRI -2026)

TapHRI: TCN-driven Touch Control of Collaborative Robots Using Only Embedded Robot Sensing

ACM/IEEE International Conference on Human-Robot Interaction (HRI -2026)

GenerativeMPC: VLM-RAG-guided Whole-Body MPC with Virtual Impedance for Bimanual Mobile Manipulation

IEEE International Conference on Systems, Man, and Cybernetics 2026

DiffusionVlog: Text-to-Aerial Cinematography

ACM/IEEE International Conference on Human-Robot Interaction (HRI -2026)

YawSitter: Modeling and Controlling a Tail-Sitter UAV with Enhanced Yaw Control

IEEE International Conference on Robotics and Biomimetics (IEEE ROBIO 2025)

Glove2UAV: A Wearable IMU-Based Glove for Intuitive Control of UAV

ACM/IEEE International Conference on Human-Robot Interaction (HRI -2026)

DroneVLA: VLA-based Aerial Manipulation

ACM/IEEE International Conference on Human-Robot Interaction (HRI -2026)

Dark Energy

Maven 18 Magazine(2018)

International Conference on Materials, Emerging Devices and Energy Efficient Technologies(2017)

Education

Bachelor of Engineering in Electronics & Communication

Anna University-St. Xavier's Catholic College of Engineering 

09.2017 – 05.2022 | Nagarcovil, India

CGPA-8.23/10(First Class).

Master of Science in Engineering Systems(Robotics)

Skolkovo Institute of Science and Technology (Skoltech) 

09.2024 – 06.2026 | Moscow, Russia

Technical Skills

Programming, AI, Robotics & Simulation:

Python, C++, C#; Isaac Sim / Lab, Unity, ROS2. ● ● ● ● ●

MATLAB & Simulink

Control System & Power Analysis ● ● ● ● ●

Equipment Expertise

Oscilloscope, Function Generator, Soldering; Digital Multimeter ● ● ● ● ●

Microsoft 365

Word, Excel, PowerPoint, and Power BI ● ● ● ● ●

Certificates

- Google Cloud Skill 
- TensorFlow Developer - DeepLearning.AI 
- MATLAB Onramp 

Courses

09.2023 – 02.2024 **Modern Robotics: Motion, Kinematics, and Dynamics Specialization** 
Northwestern University

07.2023 – 09.2023 **Deep Learning Specialization** 
Coursera-DeepLearning.AI

07.2023 – 09.2023 **Machine Learning Specialization** 
Coursera-Stanford Online

05.2023 – 07.2023 **Mathematics for ML and Data Science** 
Coursera-DeepLearning.AI

Organisations

12.2025 – Present **ACM Student Membership** 

2018 – 2020 **IEEE Student Membership**
IEEE Student Branch (62851)

2018 – 2019 **National Service Scheme**
Jal Shakti Abhiyan Award (2019) - Zonal Level National honor from the Govt. of India for an innovative water conservation concept(Eco-Green Earth).